

OC Core 1.3 / Systems / Civilizational projection Domain Packet

Status Snapshot

- Current state: `VALIDATED_ANCHOR_ACTIVE`
- Target state: `VALIDATED_ANCHOR_ACTIVE`
- Scientific class: `THEOREM_NATIVE`
- Trace status: `TRACE_COMPLETE`
- Closure verdict: `PASS`
- Quantitative pass result: `PASS`
- Next required action: `MAINTENANCE_ONLY`
- Replay status: `PASS_REPLAYABLE`

Theorem-to-Observable Map

- K8 projection claims lawfully generate the bounded systems packet over measurable macro-process trajectories and regime-shift signatures on pinned official series.
- The promoted systems packet binds source-native K8 projection structure to residual and turning-point observables without authorizing unrestricted civilizational prediction.
- Every promoted systems benchmark family is traced through the locked K8 route, official macro-process data, and held-out interval replay.

Benchmark Dataset Manifest

- `SYS_BENCH_001`: `population_trajectory_residual` | World Bank Indicators API | <https://api.worldbank.org/v2/en/indicator/SP.POP.TOTL?format=json> | held-out: leave out one interval family
- `SYS_BENCH_002`: `gdp_growth_turning_point_error` | World Bank Indicators API | <https://api.worldbank.org/v2/en/indicator/NY.GDP.MKTP.KD.ZG?format=json> | held-out: reserve one held-out macro interval
- `SYS_BENCH_003`: `process_system_regime_score` | World Bank Indicators API | <https://api.worldbank.org/v2/en/indicator/EG.USE.PCAP.KG.OE?format=json> | held-out: reserve one resource-intensity curve family

Pinned Official Route Protocol

- `SYS_ROUTE_001`: World Bank (`OFFICIAL_OPEN_DATA_API_DOCS`) -> <https://datahelpdesk.worldbank.org/knowledgebase/articles/90698-about-the-indicators-api-documentation> | purpose: Pinned macro indicator route and API semantics
- `SYS_ROUTE_002`: U.S. Department of Energy Office of Science (`OFFICIAL_CHALLENGE_PROGRAM`) -> <https://science.osti.gov/ascr> | purpose: Process-systems and high-performance systems benchmark framing

Measurable Outputs

- `trajectory_residual`
- `turning_point_error`
- `held_out_interval_regime_score`

Measurement Schema

Evaluate K8 projection packets against pinned official macro-process series with held-out interval tests.

Acceptance Criterion

Declared trajectory and regime-change observables remain inside tolerance on held-out intervals.

Falsifier

A missed turning point or out-of-band held-out trajectory falsifies the promoted systems packet.

Quantitative Acceptance Thresholds

- brier_score_max: 0.08
- coverage_ratio_required: 1.0
- critical_failure_f1_min: 0.87
- critical_failure_precision_min: 0.85
- critical_failure_recall_min: 0.9
- expected_calibration_error_max: 0.05
- policy_snapshot: {'benchmark_residual_policy': {'critical_residual_sigma_threshold': 2.5, 'schema_id': 'LOGION_BENCHMARK_RESIDUAL_POLICY_v1', 'severe_residual_sigma_threshold': 1.5}, 'prediction_accuracy_policy': {'brier_score_max': 0.08, 'cases_total_required': 30, 'coverage_ratio_required': 1.0, 'critical_failure_f1_min': 0.87, 'critical_failure_precision_min': 0.85, 'critical_failure_recall_min': 0.9, 'expected_calibration_error_max': 0.05, 'schema_id': 'LOGION_PREDICTION_ACCURACY_POLICY_v2'}, 'prediction_sigma_policy': {'cases_total_min': 30, 'normalized_error_max_max_sigma': 5.0, 'normalized_error_mean_abs_max_sigma': 1.0, 'normalized_error_p95_max_sigma': 2.5, 'schema_id': 'LOGION_PREDICTION_SIGMA_POLICY_v1', 'tail_breach_count_max': 0}}
- tail_breach_count_max: 0

Replay Harness

- Replay command: `validation/run_all.py --qa-only` (public replay umbrella; domain: SYSTEMS_CIVILIZATIONAL_PROJECTION)
- Evidence bar: HYBRID_ESCALATION
- Protocol id: OC13::EXECUTION::SYSTEMS::MACRO_TRAJECTORIES_AND_REGIME_SHIFTS

Institute-run Escalation

- Current escalation status: NOT_REQUIRED_UNLESS_NEW_OBSERVABLE_FAMILY_ENTERS
- Escalation trigger: OPEN_DATA_COVERAGE_LT_1_0_OR_HELD_OUT_INTERVAL_COUNT_LT_30_OR_TURNING_POINT
- Measurement wave id: INSTITUTE_RUN::SYSTEMS::WAVE_4A
- Measurement plan: Acquire partner or institute-run observation programs for the missing macro-process lane and rerun the held-out interval evaluation without broadening civilizational claims.

Blocking IDs

- None.